Motivation Letter: Research Engineer - Agentic Systems in AI for Science (RE2)

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Recruitment Panel
Barcelona Supercomputing Center - Centro Nacional de Supercomputación

Subject: Application for Research Engineer - Agentic Systems in AI for Science (RE2), reference 383_26_AII_AIT_RE2

Dear Members of the Recruitment Panel,

I am applying for the Research Engineer position in Agentic Systems for AI for Science because the initiative closely matches the systems problem that has defined my recent work: how to give language-model agents useful capabilities while keeping deployment, evaluation, and reported outcomes reliable. I recently completed an MSc in Modelling for Science and Engineering at the Universitat Autonoma de Barcelona, specialising in Data Science, after a BSc in Mechanical Engineering. I completed my BSc under the Hungarian Government's Stipendium Hungaricum scholarship, which covered my tuition and provided living and accommodation support. I was later selected for CSIC's JAE Intro ICU 2025 research-introduction scholarship at IIIA-CSIC, where I developed my MSc thesis. My experience combines enterprise agent workflows, interactive robotics, and computational modelling.

At Aily Labs, I took responsibility for integration seams connecting go-to-market data capabilities to SuperAgent, the company's general-purpose, customer-facing LLM reasoning agent. My work progressed from MCP server tools to task-scoped skills that packaged approved instructions, tools, and evaluation criteria around each workflow. I then helped migrate these skills into a governed semantic layer, representing approved SQL tables through curated Markdown resources with their business meaning, relationships, permitted use, and access constraints. I developed and maintained GTM capabilities used by Aily tenants including Sanofi.

When answer quality regressed or a skill, resource mapping, or workflow contract changed, my team was responsible for evaluating and correcting the behaviour from development through production. My contribution was to trace failures across data availability, semantic descriptions, table allowlists, tool selection, DAG payloads, and post-processing, then verify the correction before promotion. Code review, automated tests, continuous integration, pre-production validation, question sets, and trace inspection provided the release evidence. This gave me practical LLMOps experience with deployment controls, production observability, regression evaluation, and tenant-scoped data and capabilities.

My MSc thesis at IIIA-CSIC brought the same concerns into an embodied system. I designed and validated a modular ROS 2 and ROS4HRI architecture for NAO that separates dialogue, grounded symbolic state, language-model planning, deterministic skill execution, and feedback-driven recovery. Planning is bounded by a reviewed capability registry, execution preserves plan lineage, and completion reports must remain tied to runtime evidence. I validated the integrated stack through a scenario-based framework covering dialogue, knowledge-base cases, composite actions, injected failures, recovery, final reporting, and physical robot checks. The result is an agentic workflow whose model, harness, ROS runtime, state, and execution path can be inspected separately.

The MSc also provided formal training relevant to BSC's HPC environment. Parallel Programming covered C, CUDA, OpenMP, MPI, SLURM, GPU programming, and performance analysis. Distributed Systems covered cluster and supercomputer infrastructure, batch queues, job arrays and dependencies, heterogeneous resources, cloud systems, and fault tolerance. Through laboratory and project work at UAB and IIIA-CSIC, I submitted and monitored cluster workloads and worked with parallel and distributed execution models. This is hands-on academic HPC experience, which I am prepared to deepen within BSC's production infrastructure.

My personal projects extend the same direction. An unpublished Universal Agentic Harness proof of concept projects task-specific capabilities from a registry, applies deterministic reach and effect gates, and records append-only traces through portable environment adapters. Watson is my working personal agent platform: Hermes provides the harness and tool workflow, LiteLLM provides model routing and an OpenAI-compatible proxy, llama.cpp serves a quantised Qwen model, and ZeroTier exposes the endpoint across my machines. I maintain its launch, recovery, routing, and performance gates.

iTrader provides complementary model-side experience as an autonomous trading platform under staged development. Its implemented offline stack learns from OHLCV windows and engineered features through validated task specifications, PPO/GAE, FiLM-conditioned policy and value networks, decomposed rewards, local LLM proposers, and reproducible manifests. Its online design progresses through replay and shadow operation before live execution, with measured solver capability, deterministic decision gates, and separate risk controls.

I can contribute immediately to the review and implementation of agent frameworks, Python-based integration, reproducible evaluation, documentation, and operational diagnosis. My direct experience is with contributing to and maintaining production LLM workflows rather than operating an institutional HPC agent platform at BSC's scale. The combination of production-facing ownership at Aily, full-stack agent validation in my thesis, and formal parallel-computing training provides a relevant foundation for that next level of responsibility.

BSC's objective of hosting an in-house agentic solution for researchers is especially compelling because it treats model choice, software architecture, HPC integration, user support, and evaluation as one engineering problem. I would bring disciplined ownership, strong Python and Linux practice, fluent English and Spanish, and experience communicating across data, research, and software teams. I would welcome the opportunity to help develop a system that improves scientific productivity while remaining reproducible and observable.

Thank you for considering my application. I would welcome a discussion about how my experience in agentic systems, deployment discipline, and evidence-based evaluation could support the AI Institute's AI4Science initiative.

Yours sincerely,

Juan David Bendek Williamson

References

1. **Francisco Martin**, Principal Data Scientist, Aily Labs

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